CS342 · VISUAL PROGRAMMING

Labs Practice

لابات د. حمد الرشيدي · تدريب Output وكتابة كود · محارب

النوع	الأسئلة	من وين
Find Output	Q1 – Q5	Lab 1: Ex 1, 2, 3, 4, 5, 6
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كل الأكواد مجربة، والمخرجات في ملف الحل هي اللي تطلع فعلًا. وأخطاء الكتابة اللي في نص اللاب (مثل liste و Time والفاصلة المنقوطة الناقصة) مصلحة هنا.

Q1 **Division by zero** OUTPUT Lab 1 · Ex 5

(a) What does the program print when y = 0?

```
public class Exemple1 {
    public static void main(String[] args) {
        int x = 10, y = 0;
        int z = x / y;
        System.out.println(x + " / " + y + " = " + z);
        System.out.println("Program End");
    }
}
```

(b) What does it print after adding try / catch?

```
public class Exemple1 {
    public static void main(String[] args) {
        int x = 10, y = 0;
        try {
            int z = x / y;
            System.out.println(x + " / " + y + " = " + z);
        }
        catch (ArithmeticException e) {
            System.out.println("Error: " + e.getMessage());
        }
        System.out.println("Program End");
    }
}
```

Lab 1 · Ex 6 OUTPUT **CalculMoyenne** Q2

What does the program print for each array? Then: which predefined exceptions are caught, and which exceptions were created by the programmer?

(1) {"14", "12", "13", "16", "18", "15"}

(2) {"x", "2", "-1", "30", "18"}

(3) {"x", "-3", "2.3"}

```
class ExceptionEmpty extends Exception {
    public String toString() { return "No note is valid"; }
}
class ExceptionRange extends Exception {
    public ExceptionRange(String s) { super(s); }
}
public class CalculMoyenne {
    private static int average(String[] list) throws ExceptionEmpty {
        int sum = 0, integer, nbNotes = 0;
        for (int i = 0; i < list.length; i++) {
            try {
                integer = Integer.parseInt(list[i]);
                if (integer < 0)
                    throw new ExceptionRange("few");
                else if (integer > 20)
                    throw new ExceptionRange("large");
                else {
                    sum += integer;
                    nbNotes++;
                }
            }
            catch (NumberFormatException nfe) {
                System.out.println("Note no." + (i + 1) + " is not complete");
            }
            catch (ExceptionRange ei) {
                System.out.println("Note no." + (i + 1) + " is too " + ei.getMessage());
            }
        }
        if (nbNotes == 0)
            throw new ExceptionEmpty();
        return sum / nbNotes;
    }
```

```
public static void main(String[] args) {
    String tab[] = { ... };
    try {
        System.out.println("The average is: " + average(tab));
    }
    catch (ExceptionEmpty er) {
        System.out.println(er);
    }
}
```

Lab 1 · Ex 1 OUTPUT **Person, Customer and Owner** Q3

Find the output of the program below.

```
class Person {
    private String lastname, firstname;
    public Person(String l, String f) {
        lastname = l;
        firstname = f;
        System.out.println("Person created");
    }
    public void displaysPerson() {
        System.out.println(lastname + " " + firstname);
    }
}
class Customer extends Person {
    private int numidentity;
    public Customer(String l, String f, int n) {
        super(l, f);
        numidentity = n;
        System.out.println("Customer created");
    }
    public void displaysCustomer() {
        displaysPerson();
        System.out.println("ID: " + numidentity);
    }
}
class Owner extends Person {
    private String nameclub, adrclub;
    public Owner(String l, String f, String n, String a) {
        super(l, f);
        nameclub = n;
        adrclub = a;
        System.out.println("Owner created");
    }
    public void displaysOwner() {
        displaysPerson();
        System.out.println(nameclub + " - " + adrclub);
    }
}
public class Test {
    public static void main(String[] args) {
        Customer c = new Customer("Ali", "Ahmad", 1234);
        c.displaysCustomer();
        Owner o = new Owner("Saad", "Omar", "Star Video", "Buraydah");
        o.displaysOwner();
        Person p = c;
```

```
        p.displaysPerson();
    }
}
```

Lab 1 · Ex 3 OUTPUT **Shapes** Q4

Find the output of the program below.

```
class Point {
    private double x, y;
    public Point(double x, double y) { this.x = x; this.y = y; }
}
abstract class Shape {
    public abstract double perimeter();
    public abstract double area();
}
class Circle extends Shape {
    private Point center;
    private double radius;
    public Circle(Point c, double r) { center = c; radius = r; }
    public void setRadius(double r) { radius = r; }
    public double getRadius() { return radius; }
    public double perimeter() { return 2 * 3.14 * radius; }
    public double area() { return 3.14 * radius * radius; }
}
class Rectangle extends Shape {
    private double length, breadth;
    public Rectangle(double l, double b) { length = l; breadth = b; }
    public void setLength(double l) { length = l; }
    public double perimeter() { return 2 * (length + breadth); }
    public double area() { return length * breadth; }
}
public class TestForm {
    public static void main(String[] args) {
        Shape[] t = { new Rectangle(2, 3), new Circle(new Point(0, 0), 1) };
        for (int i = 0; i < t.length; i++)
            System.out.println(t[i].area() + " " + t[i].perimeter());
        Circle c = (Circle) t[1];
        c.setRadius(2);
        System.out.println(t[1].area());
        ((Rectangle) t[0]).setLength(5);
        System.out.println(t[0].perimeter());
    }
}
```

Lab 1 · Ex 2 – 4 OUTPUT **Compile or not** Q5

Using the lab classes below, say for each line: **OK** (and what it prints), **compile error** or **run-time error**.

```
interface Displayable { void display(); }
interface Shape { double area(); double perimeter(); }
abstract class Person { abstract void writePerson(); }
class Point implements Displayable { ... double getX() ... }   // display() prints "x,y"
class Circle implements Shape, Displayable { ... }              // display() prints "Circle r"
class Rectangle implements Shape, Displayable { ... }           // display() prints "Rect"

(a)  Person p = new Person();
(b)  Shape s = new Shape();
(c)  class Square implements Shape { public double area() { return 1; } }
(d)  Displayable d = new Point(1, 2);  d.display();
(e)  Displayable d = new Point(1, 2);  double v = d.getX();
(f)  Shape s = new Circle(1, new Point(0, 0));  Circle c = (Circle) s;  c.display();
(g)  Shape s = new Circle(1, new Point(0, 0));  Rectangle r = (Rectangle) s;
(h)  Displayable[] t = new Displayable[3];
     t[0] = new Point(1, 2);  t[1] = new Rectangle(2, 3);  t[2] = new Circle(1, (Point) t[0]);
     for (int i = 0; i < 3; i++) t[i].display();
```

Lab 1 · Ex 1 WRITE **Write Person and Customer** Q6

1) Define a Person class with two private attributes: lastname, firstname, a constructor that initializes them, and a method void displaysPerson().
2) Define a Customer class inheriting from Person with a private attribute numidentity, a constructor (lastname, firstname, numidentity) and a method void displaysCustomer().

Lab 1 · Ex 5 WRITE **Three ways to handle division by zero** Q7

Modify the program (x = 10, y = 0) so that it ends well:

1. Using a predefined exception class.
2. Using a predefined exception class with a custom message.
3. Using your own exception class (give a meaningful name and message).

Lab 1 · Ex 7 WRITE **TempsException** Q8

Modify the constructor of the class Temps so that it throws a TempsException (to create) if hours, minutes or seconds do not form a valid time. Change main so that the TempsException is handled by displaying the message "Invalid time".

```
public class Temps {
    private int hours;
    private int minutes;
    private int seconds;

    public Temps(int h, int m, int s) {
        hours = h;
        minutes = m;
        seconds = s;
    }
```

```
    public static void main(String[] args) {
        Temps t = new Temps(24, 12, 67);
    }
}
```

Lab 2 · Ex 1 WRITE **MyWindow as a class** Q9

Modify the class MyWindow to get the same GUI when the main program becomes:

`public static void main(String[] args) { MyWindow f = new MyWindow(); }`

```
import javax.swing.*;
public class MyWindow {
    public static void main(String[] args) {
        JFrame f = new JFrame("MyWindow");
        JLabel label = new JLabel("Good Morning");
        JPanel p = (JPanel) f.getContentPane();
        p.add(label);
        f.setSize(300, 200);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```

Lab 2 · Ex 2 WRITE **Sports club members (challenge)** Q10

Write the Java program that displays exactly the window below, using JLabel, JTextField, JTextArea, JRadioButton, ButtonGroup, JCheckBox, JComboBox, JList and JButton, and different layout managers.

